
Diffuse Cascade in 3D Navier-Stokes: Time-Resolved Evidence for Triad Equidistribution

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Abstract

We present time-resolved numerical evidence that triad amplitudes in the three-dimensional incompressible Navier-Stokes equations remain equidistributed across spectral shells and time. Across resolutions $N=32, 64, 128$, no concentration events are observed under stringent diagnostics. The minimum amplitude equidistribution (AET) remains above 0.308, the scaling exponent remains near zero, and no helical sector exhibits persistent dominance. These results indicate a diffuse cascade in which energy transfer is continuously distributed across triads, with no transient or sustained concentration in the tested regimes.

1. Introduction

The problem of global regularity for the three-dimensional incompressible Navier-Stokes equations hinges on whether nonlinear interactions can concentrate energy into small subsets of modes or triads, potentially leading to finite-time blowup. This work investigates that possibility directly by analyzing the structure and distribution of triad interactions in spectral space. Rather than bounding norms indirectly, we examine whether energy transfer concentrates or remains distributed across interacting triads over time.

2. Methodology

We use a pseudo-spectral Navier-Stokes solver with triad-level diagnostics. For each snapshot and spectral shell j , we compute:

- Amplitude Equidistribution: $AET_j(t) = N_{eff,j}(t) / N_j$
- Scaling exponent: $AET_j(t) \sim C(t) \cdot 2^{-\alpha(t) \cdot j}$
- Concentration diagnostics: top 1% triad mass fraction, helical sector dominance, time-persistent collapse detection

A "danger event" is defined as any snapshot where $\text{AET}_j(t) < 0.05$, top 1% mass $> 30\%$, or $\alpha(t) > 1$.

3. Results

Resolution Summary

Resolution	Danger Events	max $\alpha(t)$	min AET(t)
N = 32	0	0.187	0.316
N = 64	0	0.061	0.319
N = 128	0	0.066	0.308

Observed Properties

- No concentration events observed across any resolution
- Top 1% triad mass $\leq 10.6\%$
- No persistent helical sector dominance
- AET remains stable across shells and time
- Scaling exponent remains near zero

4. Time Evolution

At all resolutions, the system exhibits stable AET across all snapshots, no transient collapse events, no sustained concentration behavior, and no shell-level drift toward localization. The scaling exponent $\alpha(t)$ remains near zero at all times, indicating that equidistribution persists across spectral shells dynamically — not just on average.

5. Interpretation

These results demonstrate that triad amplitudes remain broadly distributed, no subset of triads carries dominant energy transfer, no coherent structure emerges to sustain concentration, and energy is continuously redistributed across interactions. In particular: energy transfer remains distributed across triads at all times. No transient or sustained concentration mechanism is detected.

6. Discussion

The observed behavior is consistent with a diffuse cascade, in which energy transfer occurs through a distributed network of interactions rather than concentrated pathways. This challenges the viability of blowup mechanisms that rely on triad-level concentration, persistent alignment, or localized amplification. Instead, the cascade appears structurally resistant to concentration.

7. Limitations

These results are based on finite-resolution simulations. While consistent across multiple resolutions $N=32, 64, 128$, the asymptotic behavior as $j \rightarrow \infty$ remains an open question requiring higher-resolution studies.

8. Conclusion

We provide numerical evidence that triad amplitudes remain equidistributed across shells and time, no concentration events occur across tested resolutions, and energy transfer is dynamically distributed rather than localized. This supports a picture of the Navier-Stokes cascade as a diffuse, non-concentrating system within the tested regimes.